



ECCOMAS

3rd - 7th June 2024
Lisbon

9th European Congress
on Computational
Methods In Applied
Sciences and
Engineering

REDUCED ORDER MODELLING VIA PGD FOR THE 3D PRINTING OF PATIENT-SPECIFIC ANATOMIES

Loïc Guezou^{1,†}

Bing Tie[†], Andrea Barbarulo[†]

[†]Université Paris-Saclay, ENS Paris-Saclay, CentraleSupélec, CNRS,
LMPS - Laboratoire de Mécanique Paris-Saclay.

¹ loic.guezou@centralesupelec.fr

- ◆ **Introduction**
- ◆ **Model order reduction using PGD with the loading position as a parameter**
 - ◆ An introductory example using a linear elastic material
 - ◆ Extension of the method to a hyperelastic case
 - ◆ *The PGD equations*
 - ◆ *The algorithm dealing with the non-linearity induced by the material model*
- ◆ **Conclusions & perspectives**

Introduction

The EndoVX project

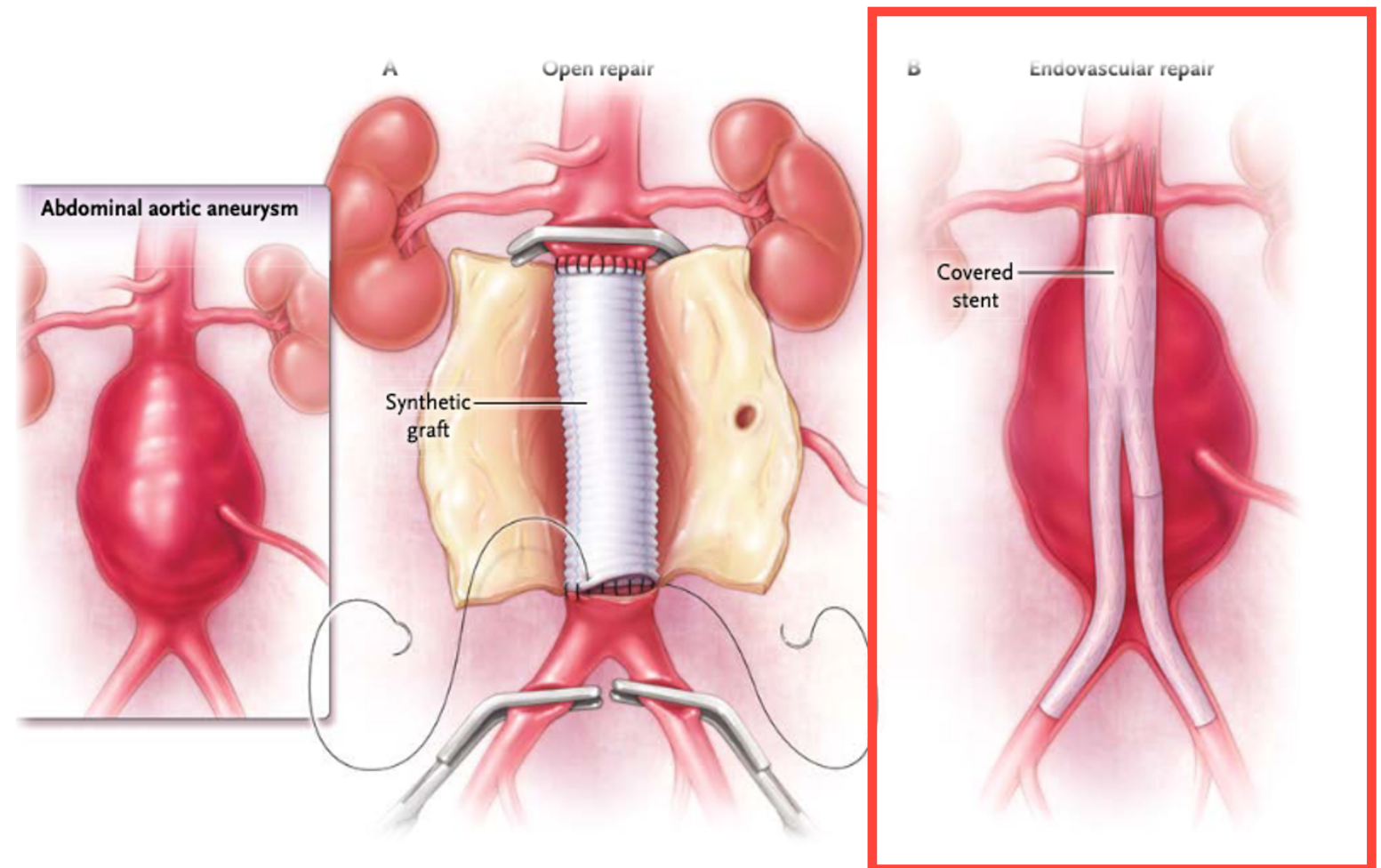
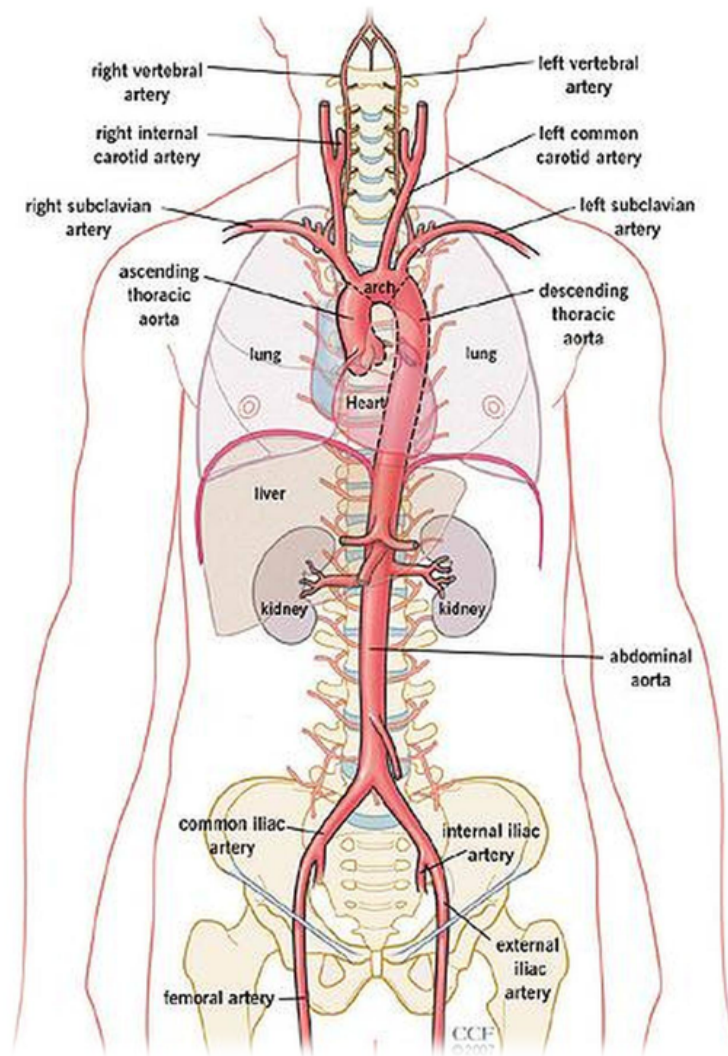
- **Multi-partners project** (CNRS, CentraleSupélec, Groupe Hospitalier Paris Saint Joseph ...)
- New technologies for a fast and personalized **treatment of aortic lesions**
- **PIA-RHU** (ANR - Recherche Hospitalo-Universitaire) funding
- Improve the **endovascular management of aortic lesions**, in particular of **complex aortic aneurysms**
- Work by a **multidisciplinary team** (AI, digital simulation, 3D printing...)



GROUPE
HOSPITALIER
PARIS
SAINT-JOSEPH



Quick background on aortic aneurysms



Aneurysm: permanent localized dilatation of an artery having at least a 50% increase in diameter. (Johnston et al. 1991)

“**Endovascular repair** [...] : intraluminal introduction of a covered stent through the femoral and iliac arteries; the stent functions as a sleeve that passes through the aneurysm sac, anchoring in the normal aorta above the aneurysm and in the iliac arteries below the aneurysm.” (Kent 2014)

General objectives

- **Overarching objective:** creating a numerical model for parametric optimization of biomimetic anatomies

DICOM files

Numerical optimization

3DP material files

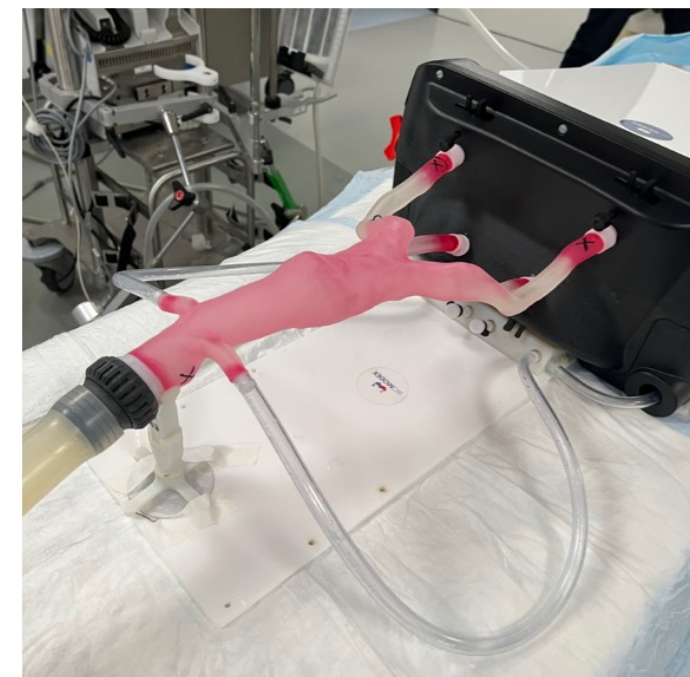
Printed model

Hyperelastic material modelling
to mimic more accurately bio/3DP material

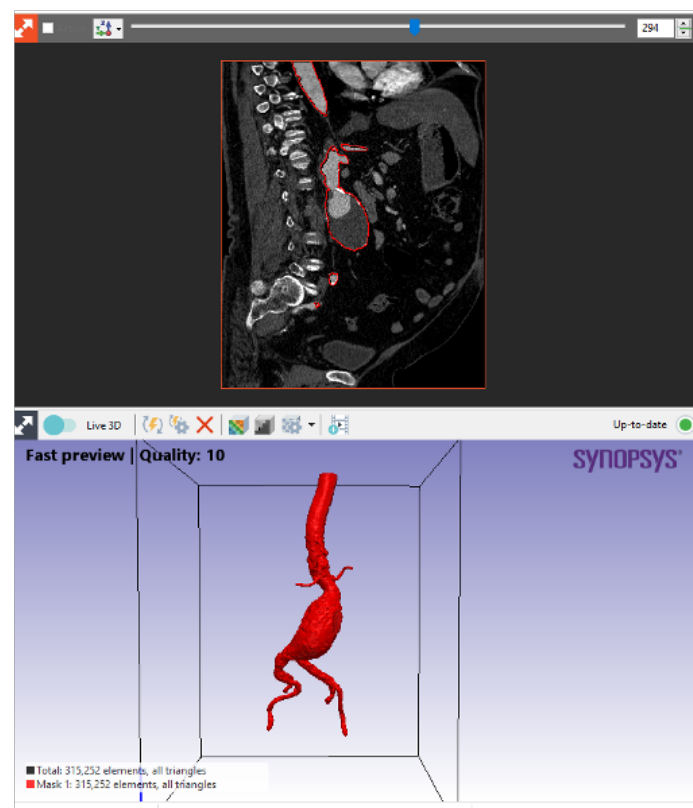
Model order reduction (PGD)
for efficient computation/separation tbd

$$\begin{bmatrix} \text{Yellow} & \text{Blue} & - & \text{Yellow} \end{bmatrix} = 0$$

$$\text{Yellow}^T \begin{bmatrix} \text{Yellow} & \text{Yellow} & - & \text{Yellow} \end{bmatrix} = 0$$



Operation simulation setup
with the printed model
[author's collection, 2022]



DICOM files from a scan of the
aorta [Biomodex, 2022]

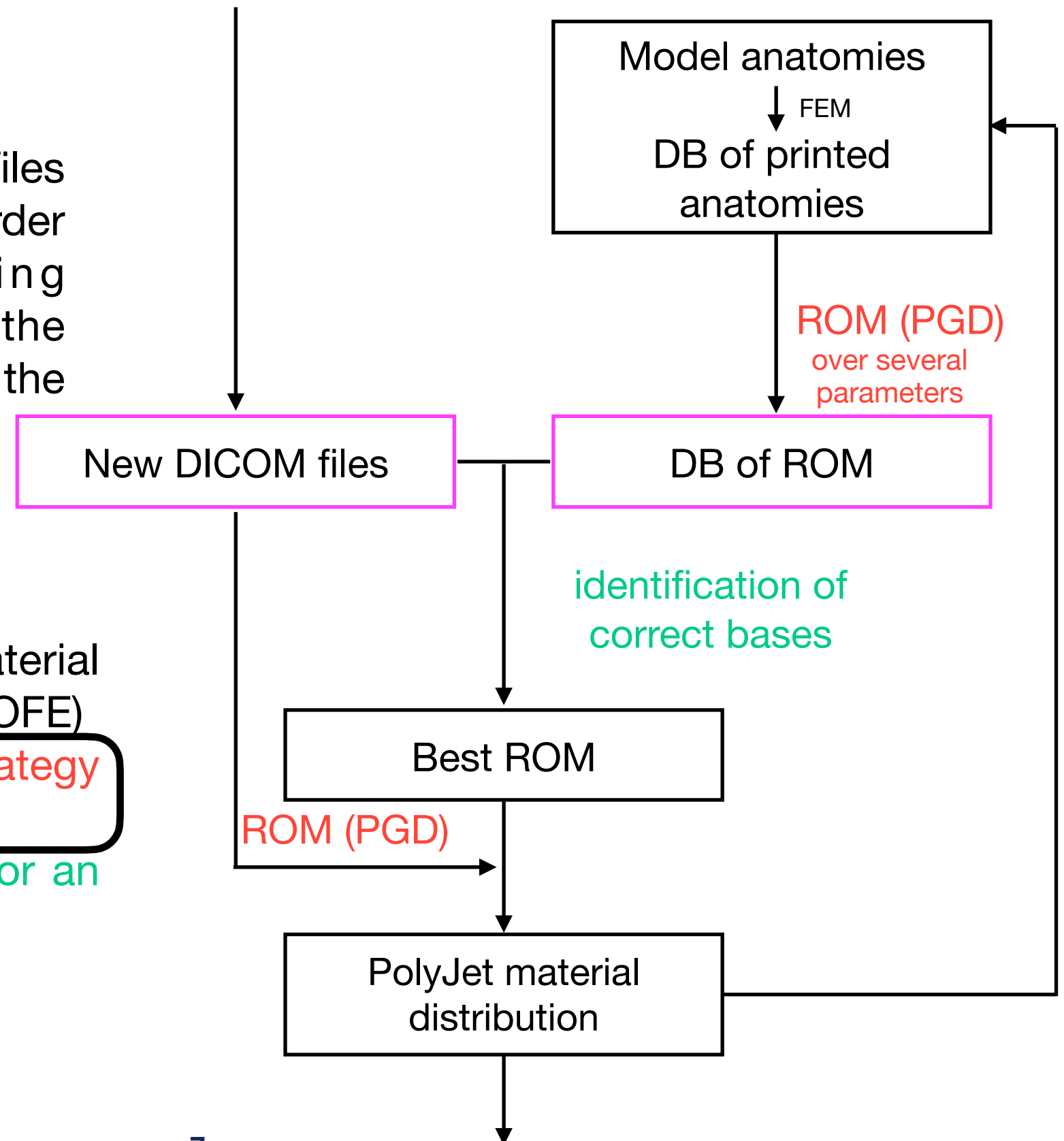
General objectives

Main goal

A tool to correlate the DICOM files with a database of reduced order models, with several varying parameters to obtain directly the correct material distribution for the model to be printed

3 steps

1. Implementing hyperelastic material models in our in-house code (OOFE)
2. Developing the ROM (PGD) strategy adapted to our formulations
3. Identification of correct bases for an existing model (optimization)



PGD using loading position as a parameter

Introductory example

Linear elastic case

Model problem

Considering an undeformed continuum body Ω bounded by $\partial\Omega$
 $\partial_u\Omega$ and $\partial_\sigma\Omega$ partition of $\partial\Omega$

$$\begin{aligned}\nabla \cdot \underline{\underline{\sigma}} + \mathbf{f} &= \mathbf{0} && \text{on } \Omega \\ \mathbf{u} &= \bar{\mathbf{u}} && \text{on } \partial_u\Omega \\ \underline{\underline{\sigma}} \cdot \mathbf{n} &= \mathbf{g} && \text{on } \partial_\sigma\Omega\end{aligned}$$

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Variational formulation

$$a(\mathbf{u}, \mathbf{u}^*) = l(\mathbf{u}^*)$$

$$\begin{aligned}a(\mathbf{u}, \mathbf{u}^*) &= \int_{\Omega} \underline{\underline{\varepsilon}}(\mathbf{u}) : \mathbb{C} : \underline{\underline{\varepsilon}}(\mathbf{u}^*) \, d\Omega \\ l(\mathbf{u}^*) &= \int_{\Omega} \mathbf{f} \cdot \mathbf{u}^* \, d\Omega + \int_{\partial_\sigma\Omega} \mathbf{g} \cdot \mathbf{u}^* \, d\partial\Omega\end{aligned}$$

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Parameter definition

Loading position $s \in \Gamma_s$.

Loading type:

- Volume force $\mathbf{f} \implies \Gamma_s \subset \Omega$
- Surface force $\mathbf{g} \implies \Gamma_s \subset \partial\Omega$

PGD decomposition

$$\mathbf{u}^n(\mathbf{x}, s) = \sum_{i=1}^n \phi^i(\mathbf{x}) \lambda^i(s) \text{ for } n \in [1, N_{mod}]$$

Variational formulation

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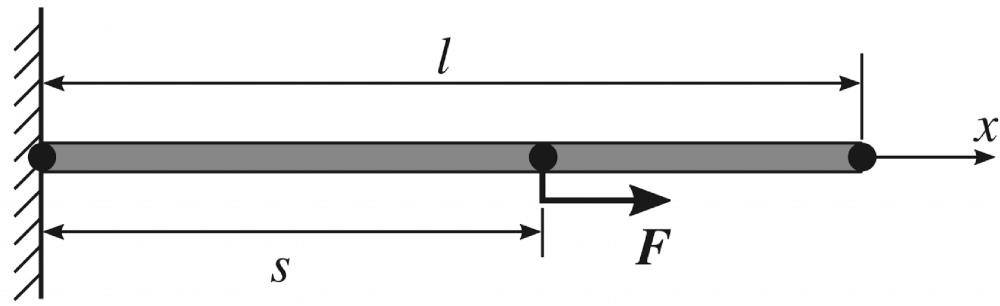
$$l(\mathbf{u}^*) = \int_{\Omega} \mathbf{f} \cdot \mathbf{u}^* \, d\Omega + \int_{\partial_\sigma\Omega} \mathbf{g} \cdot \mathbf{u}^* \, d\partial\Omega$$

Extended PGD variational formulation

$$\mathcal{A}(\mathbf{u}, \mathbf{u}^*) = \mathcal{L}(\mathbf{u}^*)$$

$$\mathcal{A}(\mathbf{u}, \mathbf{u}^*) = \int_{\Gamma_s} \int_{\Omega} \underline{\underline{\varepsilon}}(\mathbf{u}) : \mathbb{C} : \underline{\underline{\varepsilon}}(\mathbf{u}^*) \, d\Omega \, d\Gamma$$

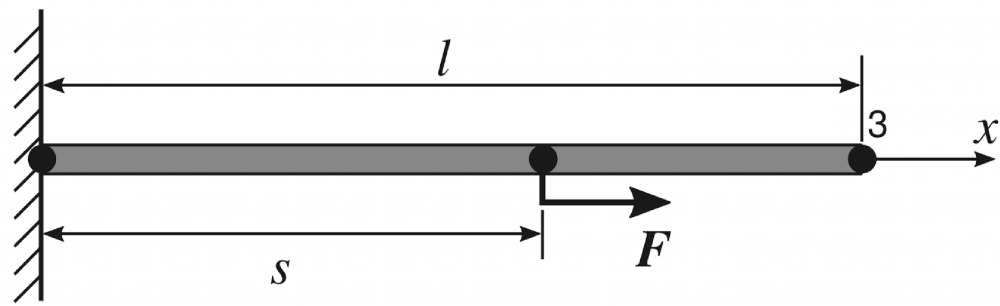
$$\mathcal{L}(\mathbf{u}^*) = \int_{\Gamma_s} \int_{\Omega} \mathbf{f} \cdot \mathbf{u}^* \, d\Omega \, d\Gamma + \int_{\Gamma_s} \int_{\partial_\sigma\Omega} \mathbf{g} \cdot \mathbf{u}^* \, d\partial\Omega \, d\Gamma$$



Sketch of the 1D model problem [Zou et al., 2018]

1D example

Linear elastic case - Zou et al., 2018



Sketch of the 1D model problem [Zou et al., 2018]

1D example

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Model problem

$$\Omega = [0, l] \text{ and } \Gamma_s = [s_1, s_2]$$

$$\frac{d}{dx} \left(EA \frac{du}{dx} \right) + f = 0$$

$$u(0) = 0$$

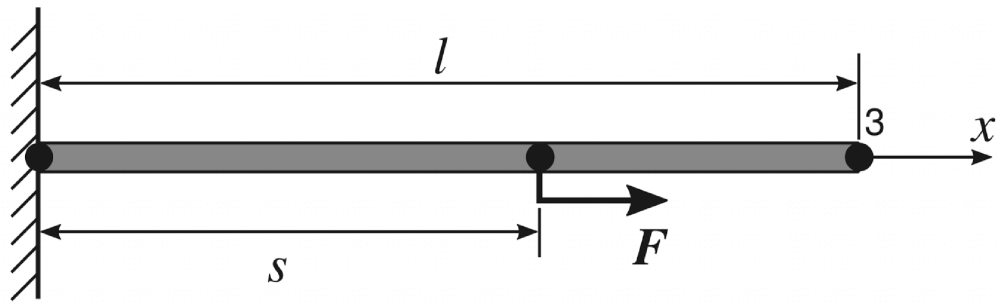
$$EA \frac{du}{dx}(l) = 0$$

$$f = F \delta(x - s)$$

Decomposition & discretization

$$u^n(x, s) = \sum_{i=1}^n \phi^i(x) \lambda^i(s) \quad \begin{aligned} \phi(x) &= \mathbf{N}^T(x) \cdot \Phi \\ \lambda(s) &= \mathbf{N}^{\lambda T}(s) \cdot \Lambda \end{aligned}$$

for $n \in [1, N_{mod}]$



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$$\text{for } n \in [1, N_{mod}]$$

$$\phi(x) = \mathbf{N}^T(x) \cdot \Phi$$

$$\lambda(s) = \mathbf{N}^{\lambda T}(s) \cdot \Lambda$$

Enrichment step: fixed point algorithm

At iteration k computing mode n ($\in [1, N_{mod}]$)

Step 1: Computing $\phi^{n(k)}$ using $\lambda^{n(k-1)}$ and u^{n-1}

$$\left(\Lambda^{n(k-1)T} \underline{\underline{M}}^\lambda \Lambda^{n(k-1)} \right) \cdot EA \underline{\underline{K}} \Phi^{n(k)} = F \underline{\underline{M}} \Lambda^{n(k-1)} - \sum_{i=1}^{n-1} \left(\Lambda^{n(k-1)T} \underline{\underline{M}}^\lambda \Lambda^i \right) \cdot EA \underline{\underline{K}} \Phi^i$$

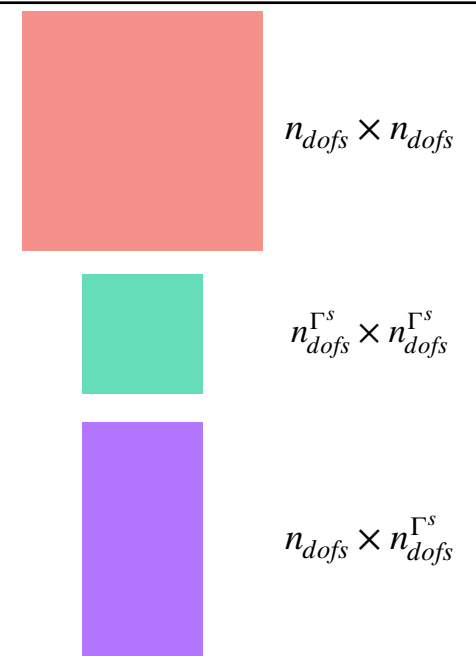
Step 2: Computing $\lambda^{n(k)}$ using $\phi^{n(k)}$ and u^{n-1}

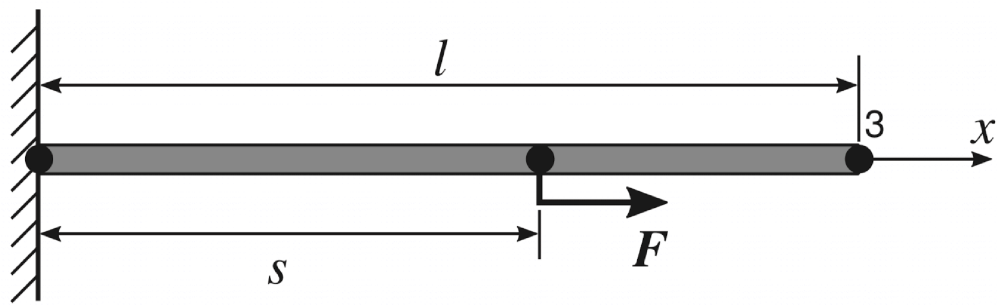
$$\left(\Phi^{n(k)T} EA \underline{\underline{K}} \Phi^{n(k)} \right) \cdot \underline{\underline{M}}^\lambda \Lambda^{n(k)} = F \underline{\underline{M}}^T \Phi^{n(k)} - \sum_{i=1}^{n-1} \left(\Phi^{n(k)T} \underline{\underline{K}} \Phi^i \right) \cdot \underline{\underline{M}}^\lambda \Lambda^i$$

$$\underline{\underline{K}} = \int_0^l \frac{d\mathbf{N}}{dx} \frac{d\mathbf{N}^T}{dx} dx$$

$$\underline{\underline{M}}^\lambda = \int_{s_1}^{s_2} \mathbf{N}^\lambda \mathbf{N}^{\lambda T} ds$$

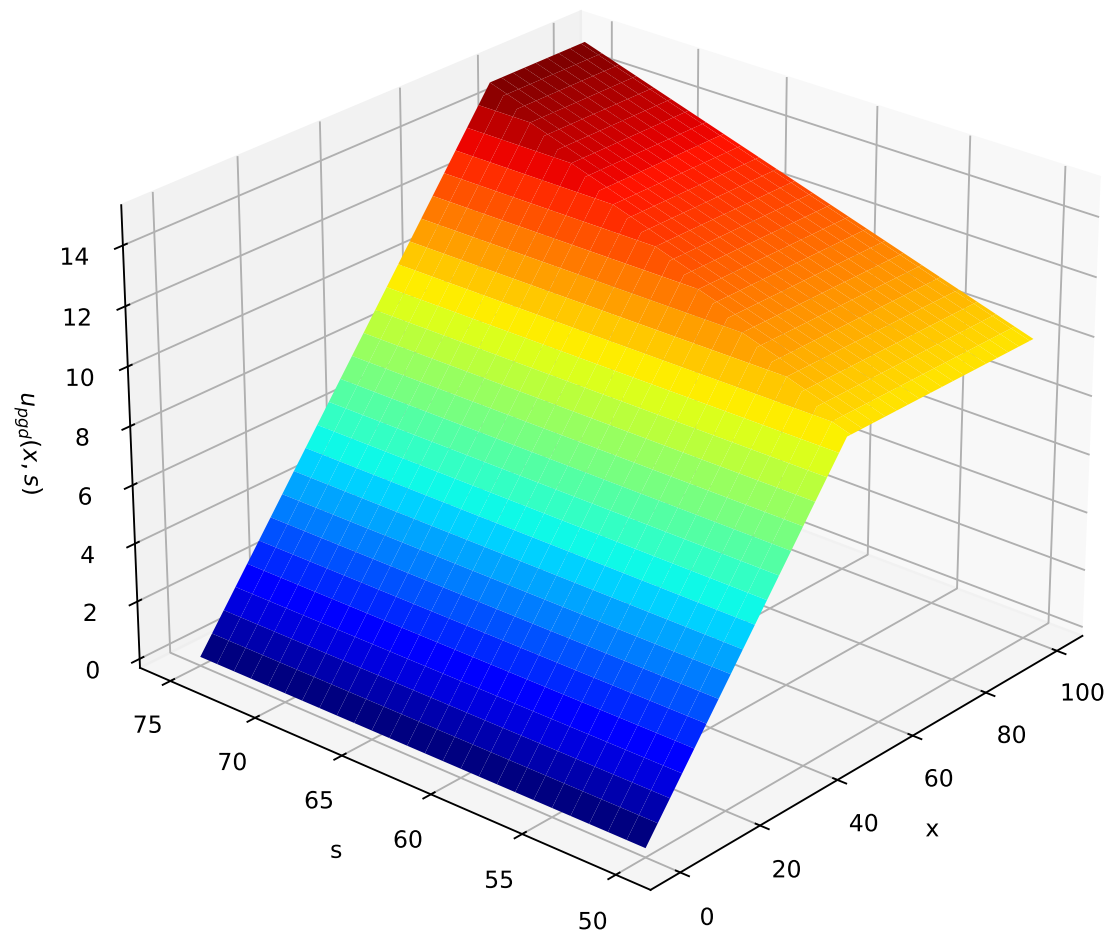
$$\underline{\underline{M}} = \int_{s_1}^{s_2} \mathbf{N}(s) \mathbf{N}^{\lambda T} ds$$





Sketch of the 1D model problem [Zou et al., 2018]

PGD solution with 30 modes - material linear_elastic

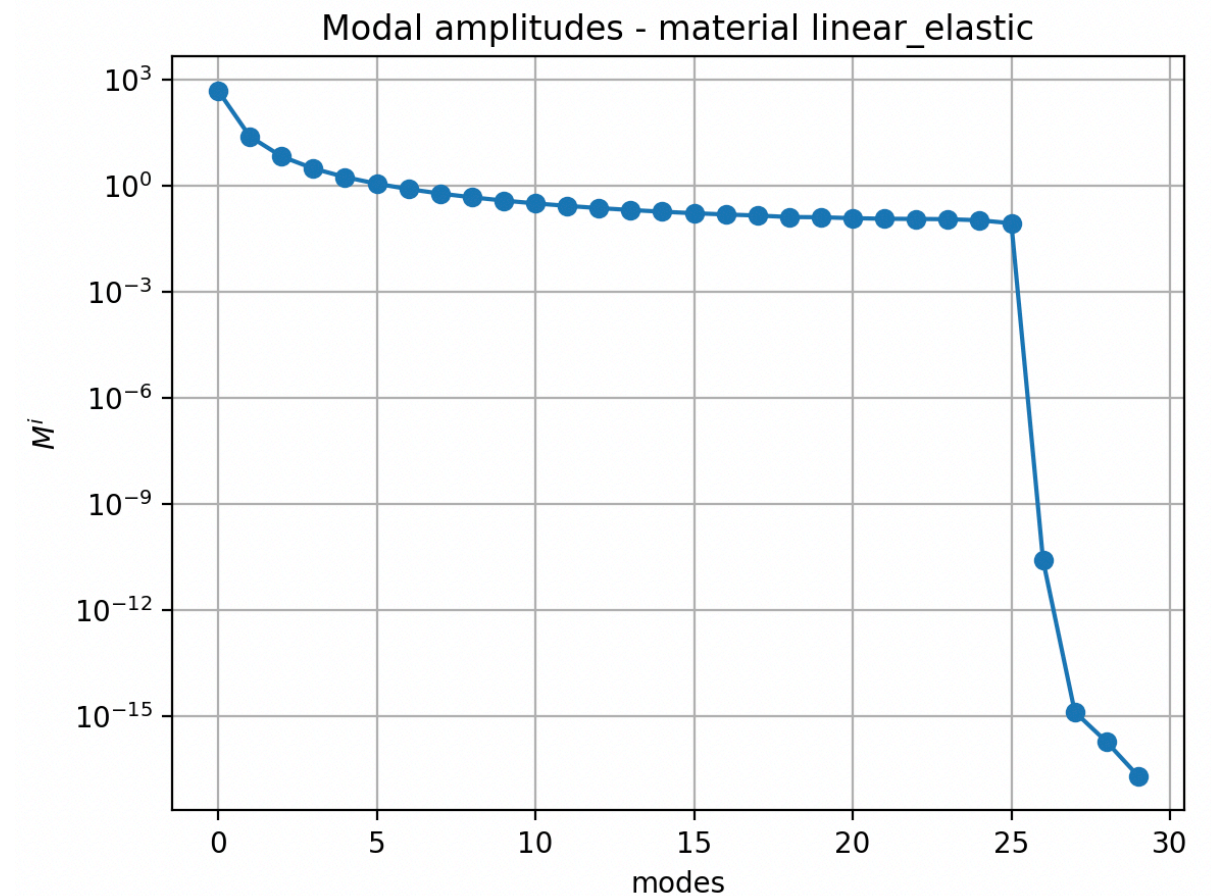


Order of magnitude of error relative to exact analytical solution:
 10^{-12}

$$err = \frac{\|u_{PGD} - u_{ana}\|}{\|u_{ana}\|} \quad u_{ana}(x, s) = \begin{cases} \frac{Fx}{EA} & \text{for } x < s \\ \frac{Fs}{EA} & \text{for } x \geq s \end{cases}$$

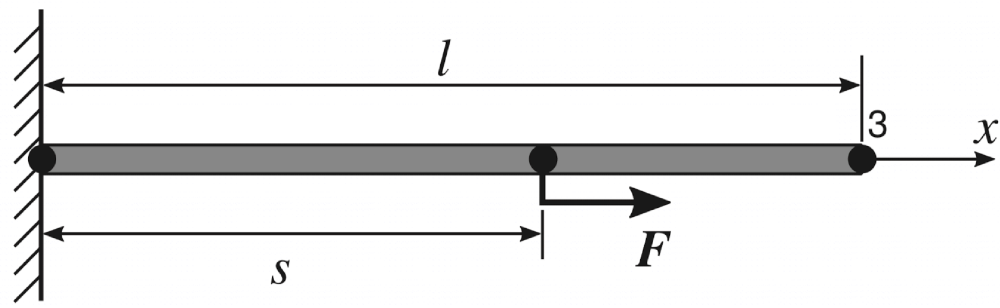
Results

Linear elastic case - Zou et al., 2018



$$\forall n \in [1, N_{mod}] \quad M^n = \|\phi^n\| \cdot \|\lambda^n\|$$

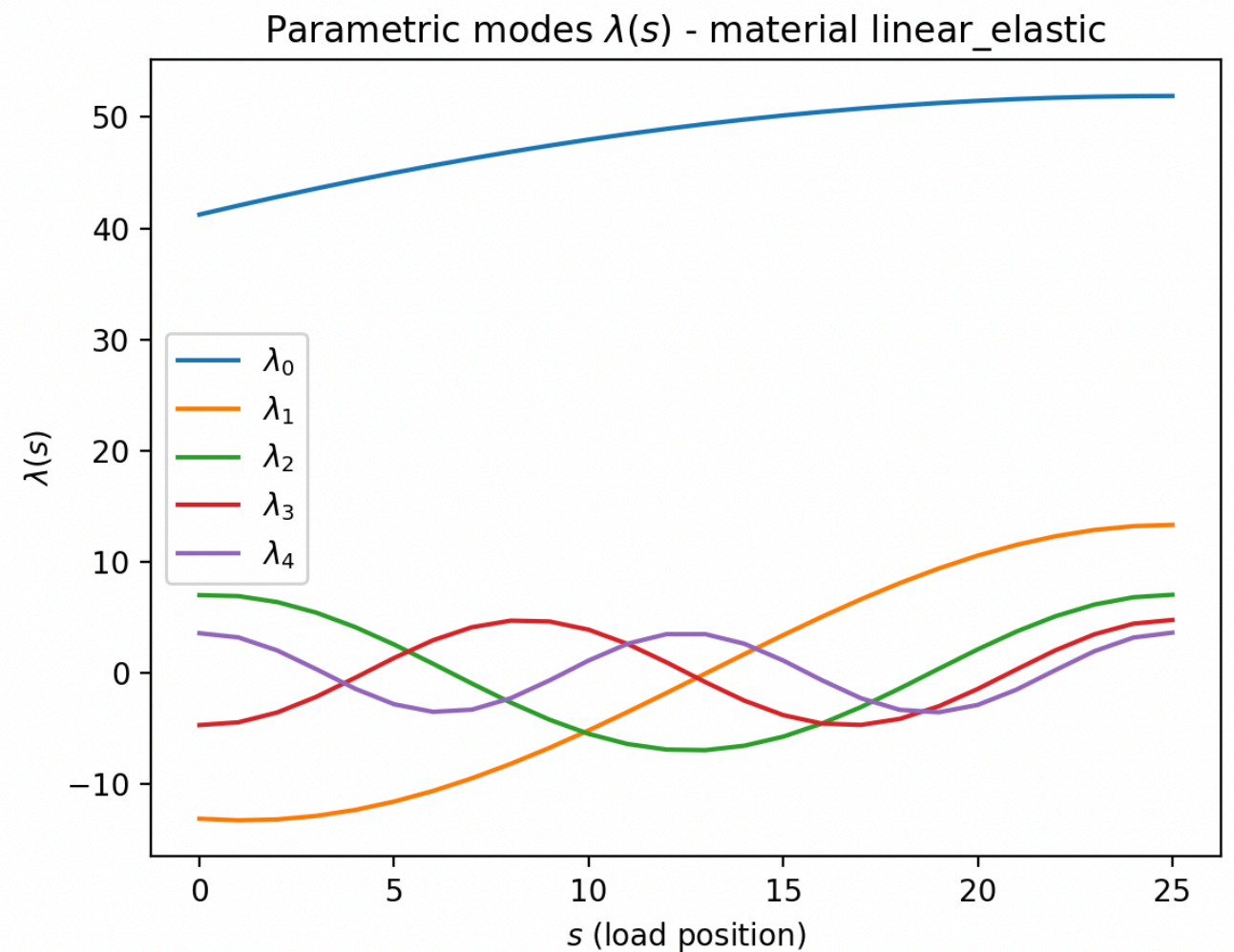
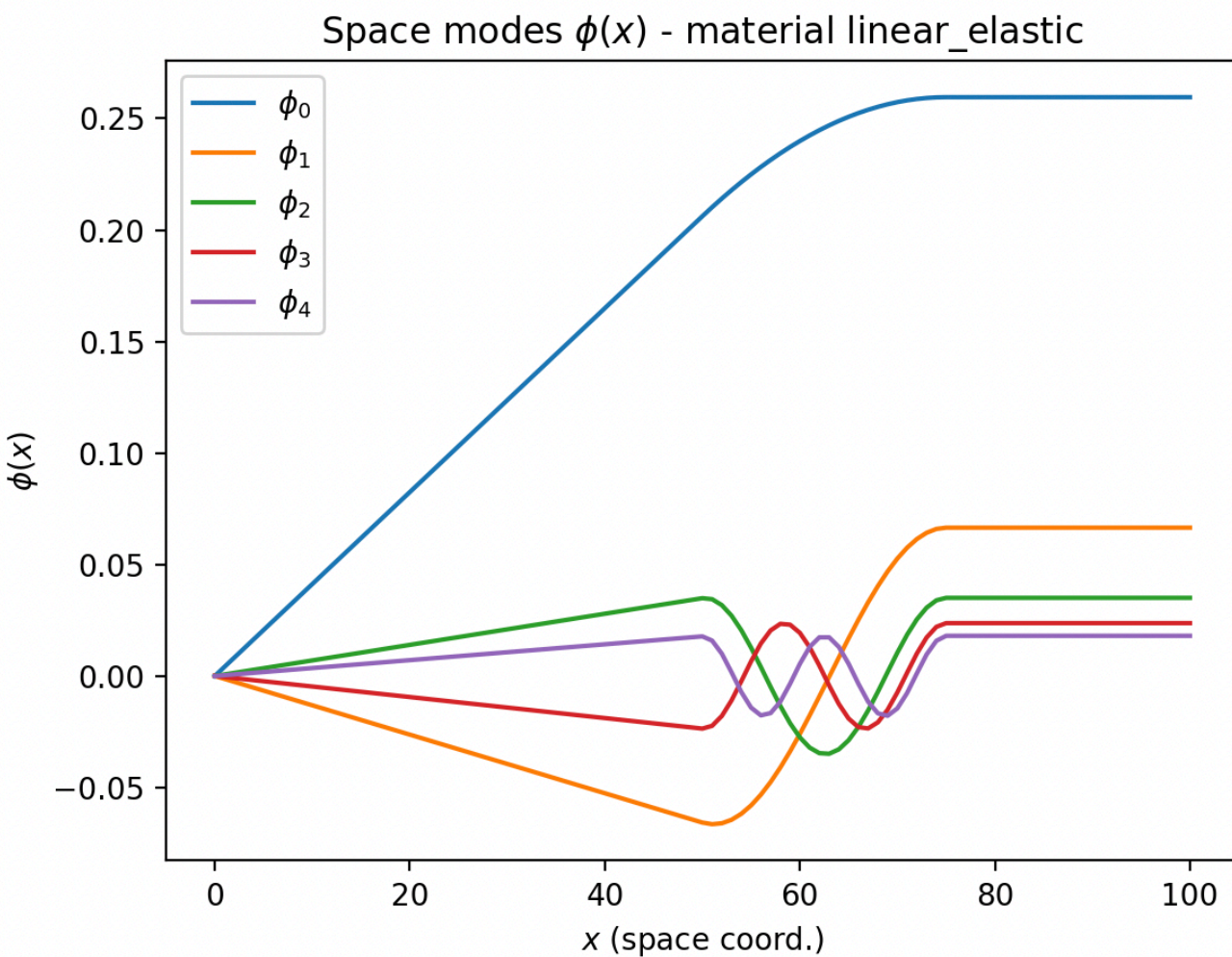
Drop in amplitude after 26th modes: agreeing with the discretization of the parametric space



Sketch of the 1D model problem [Zou et al., 2018]

Results

Linear elastic case - Zou et al., 2018



First 5 space $\{\phi^n\}_{n=0,\dots,4}$ and parametric $\{\lambda^n\}_{n=0,\dots,4}$ modes

Extension of the method

Hyperelastic case

Hyperelastic material:

Characterized by a strain energy function ψ
 Soft elastomeric materials
 Compressible, incompressible, near incompressible

Constitutive equation linking stress and strain:

$$\mathbf{S} = \frac{\partial \psi}{\partial \mathbf{C}} = 2 \frac{\partial \psi}{\partial \mathbf{E}}$$

Where:

- $\mathbf{F} = \mathbf{I} + D_x \mathbf{u}$ (transformation gradient tensor)
- $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ (right Cauchy-Green tensor)
- $\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I})$ (Euler-Lagrange strain tensor)
- $\mathbf{S}$ (2nd Piola-Kirchhoff stress tensor)

Rivlin's law (invariants-based laws)

$$\psi(\mathbf{C}) = \sum_{(p,q)=0}^{\infty} C_{pq} (I_1(\mathbf{C}) - 3)^p (I_2(\mathbf{C}) - 3)^q$$

$I_i(\mathbf{C})_{(i=1,2,3)}$ main invariants of $\mathbf{C}$

	p		q
	1	2	1
	C10	C20	C01
Signorini	✓	✓	✓
Mooney-Rivlin	✓	✗	✓
Neo-Hooke	✓	✗	✗

+ Saint Venant-Kirchhoff for simplicity

Extension of the method

Hyperelastic case

Changes compared to linear elasticity

$$\begin{aligned}\nabla \cdot \underline{\underline{P}} + \mathbf{f} &= \mathbf{0} && \text{on } \Omega \\ \mathbf{u} &= \bar{\mathbf{u}} && \text{on } \partial_u \Omega \\ \underline{\underline{P}} \cdot \mathbf{n} &= \mathbf{g} && \text{on } \partial_\sigma \Omega\end{aligned}$$



$$a(\mathbf{u}, \mathbf{u}^*) = \int_{\Omega} \underline{\underline{S}} : \underline{\underline{E}}^* \, d\Omega$$



$$\mathcal{A}(\mathbf{u}, \mathbf{u}^*) = \int_{\Gamma_s} \int_{\Omega} \boxed{\underline{\underline{S}} : \underline{\underline{E}}^*} \, d\Omega \, d\Gamma$$

nonlinearity

Extension of the method

Hyperelastic case

How to deal with nonlinearity?

Changes compared to linear elasticity

$$\begin{aligned}\nabla \cdot \underline{\underline{P}} + \mathbf{f} &= \mathbf{0} && \text{on } \Omega \\ \mathbf{u} &= \bar{\mathbf{u}} && \text{on } \partial_u \Omega \\ \underline{\underline{P}} \cdot \mathbf{n} &= \mathbf{g} && \text{on } \partial_\sigma \Omega\end{aligned}$$

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nonlinearity

Algorithm from Favoretto et al., 2019 & Fernandes et al., 2021 - *flexible and lowers computational cost*
Applied on NL heat eqn. and NS with convection

Extension of the method

Hyperelastic case

Changes compared to linear elasticity

$$\begin{aligned}
 &\underline{\nabla} \cdot \underline{\underline{P}} + \underline{\underline{f}} = \underline{\underline{0}} \quad \text{on } \Omega \\
 &\underline{\underline{u}} = \underline{\underline{\bar{u}}} \quad \text{on } \partial_u \Omega \\
 &\underline{\underline{P}} \cdot \underline{\underline{n}} = \underline{\underline{g}} \quad \text{on } \partial_\sigma \Omega \\
 &\downarrow \\
 &a(\underline{\underline{u}}, \underline{\underline{u}}^*) = \int_{\Omega} \underline{\underline{S}} : \underline{\underline{E}}^* \, d\Omega \\
 &\downarrow \\
 &\mathcal{A}(\underline{\underline{u}}, \underline{\underline{u}}^*) = \int_{\Gamma_s} \int_{\Omega} \boxed{\underline{\underline{S}} : \underline{\underline{E}}^*} \, d\Omega \, d\Gamma \quad \text{nonlinearity}
 \end{aligned}$$

How to deal with nonlinearity?

Algorithm from Favoretto et al., 2019 & Fernandes et al., 2021 - *flexible and lowers computational cost*
Applied on NL heat eqn. and NS with convection

Introduction of two linearizations:

- At mode $n = 0$ (and potentially a few select modes): « incremental linearization »
- At all the other modes: « explicit linearization »

Explicit linearization	Incremental linearization
located out of the fixed point loop, in the PGD loop	located in the fixed point loop
takes into account solution from the previous modes only	takes into account solution from the previous modes as well as the previous fixed point solution
Low computational cost	High computational cost

$$\underline{\underline{u}}^{n-1} + (\phi^{n(k-1)}, \lambda^{n(k-1)})$$

Linearizations

Hyperelastic case

Explicit linearization

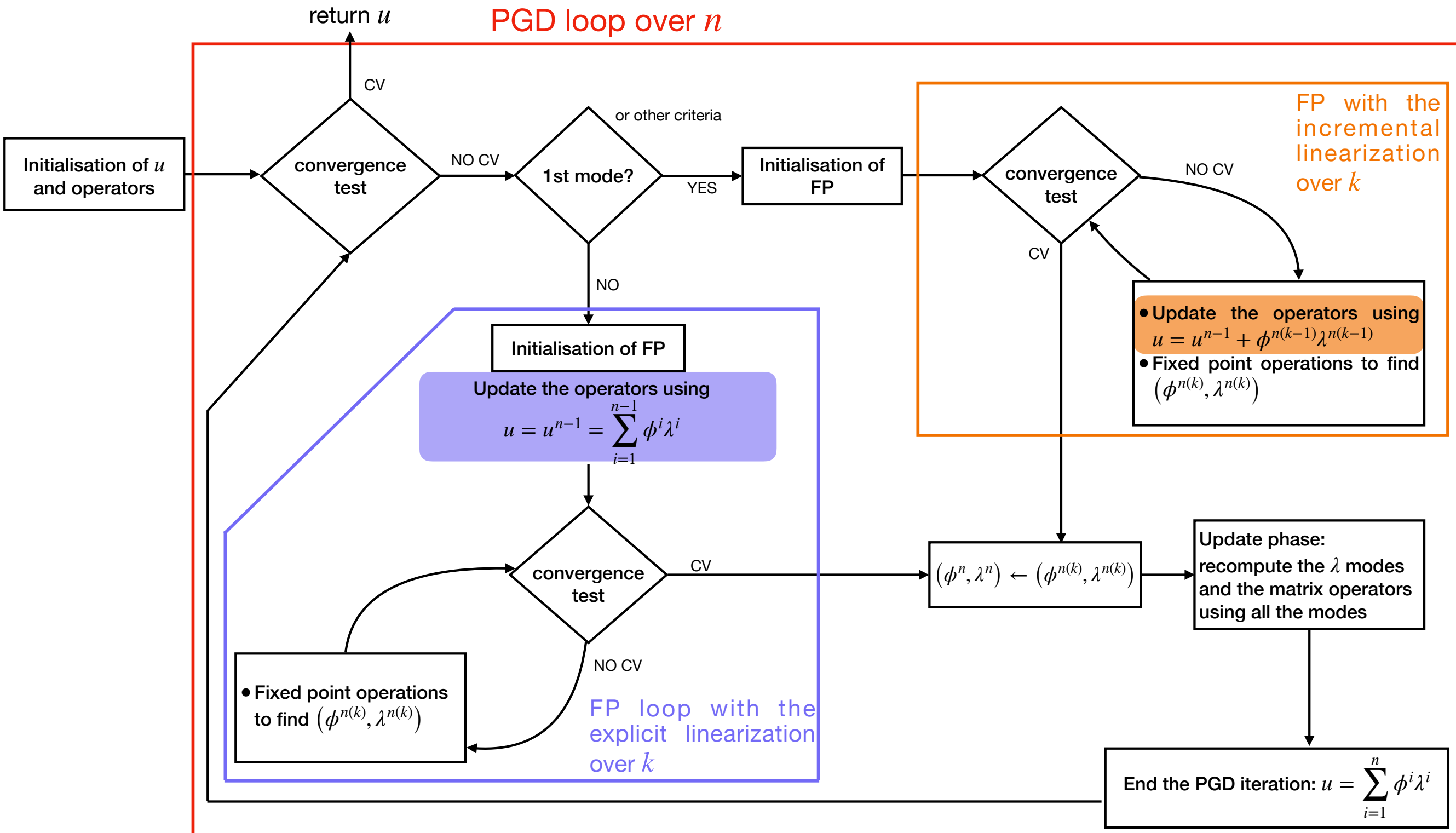
$$\begin{aligned}
 & \int_{\Gamma_s} \int_{\Omega} \left(\underline{\underline{S}}_{n-1} \cdot \nabla \mathbf{u}^n \right) : \nabla \mathbf{u}^* + \left(\underline{\underline{F}}_{n-1}^T \cdot \nabla \mathbf{u}^* \right) : \mathbb{C} : \nabla \mathbf{u}^n \, d\Omega \, d\Gamma \\
 &= \int_{\Gamma_s} \int_{\Omega} \mathbf{f} \cdot \mathbf{u}^* \, d\Omega \, d\Gamma + \int_{\Gamma_s} \int_{\partial_{\sigma} \Omega} \mathbf{g} \cdot \mathbf{u}^* \, d\partial\Omega \, d\Gamma \\
 &- \int_{\Gamma_s} \int_{\Omega} \underline{\underline{S}}_{n-1} : \underline{\underline{E}}_{n-1}^* \, d\Omega \, d\Gamma
 \end{aligned}$$

Incremental linearization

$$\begin{aligned}
 & \int_{\Gamma_s} \int_{\Omega} \left(\underline{\underline{S}} \left[\mathbf{u}^{n-1} + \phi^{n(k-1)} \lambda^{n(k-1)} \right] \cdot \nabla \mathbf{u}^n \right) : \nabla \mathbf{u}^* + \left(\underline{\underline{F}}^T \left[\mathbf{u}^{n-1} + \phi^{n(k-1)} \lambda^{n(k-1)} \right] \cdot \nabla \mathbf{u}^* \right) : \mathbb{C} : \nabla \mathbf{u}^n \, d\Omega \, d\Gamma \\
 &= \int_{\Gamma_s} \int_{\Omega} \mathbf{f} \cdot \mathbf{u}^* \, d\Omega \, d\Gamma + \int_{\Gamma_s} \int_{\partial_{\sigma} \Omega} \mathbf{g} \cdot \mathbf{u}^* \, d\partial\Omega \, d\Gamma \\
 &- \int_{\Gamma_s} \int_{\Omega} \underline{\underline{S}} \left[\mathbf{u}^{n-1} + \phi^{n(k-1)} \lambda^{n(k-1)} \right] : \underline{\underline{E}}^* \left[\mathbf{u}^{n-1} + \phi^{n(k-1)} \lambda^{n(k-1)} \right] \, d\Omega \, d\Gamma
 \end{aligned}$$

Overview of the algorithm

Hyperelastic case



Conclusions & perspectives

Global aim of the approach

Improving the 3D printed models replicating patient-specific aneurysmal aortae

- *Foundation*
 - Choice of the appropriate hyperelastic material model and implementation of the hyperelastic model in in-house FEM code (OOFE)

Conclusions & perspectives

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Improving the 3D printed models replicating patient-specific aneurysmal aortae

- *Foundation*
 - Choice of the appropriate hyperelastic material model and implementation of the hyperelastic model in in-house FEM code (OOFE)
- *Recently completed*
 - Theory for the linear elastic and hyperelastic PGD method using loading position as an extra parameter
 - Application of the method for linear elasticity to a 1D example: *good results* (as expected)

Conclusions & perspectives

Global aim of the approach

Improving the 3D printed models replicating patient-specific aneurysmal aortae

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- Choice of the appropriate hyperelastic material model and implementation of the hyperelastic model in in-house FEM code (OOFE)

- *Recently completed*

- Theory for the linear elastic and hyperelastic PGD method using loading position as an extra parameter
- Application of the method for linear elasticity to a 1D example: *good results*



What's coming

- ✦ Numerical implementation of the PGD strategy for hyperelasticity using the innovative algorithm
- ✦ Fast, flexible way of dealing with nonlinearity in PGD

**Thank you for your
attention**